

A Little- o and Big- O Notation

Definition 1 (Little- o Notation). Let f and g be real-valued functions defined on a neighborhood of 0. The function $f(h)$ is said to be $o(g(h))$ as $h \rightarrow 0$ if $\forall \epsilon > 0 \exists \delta > 0$ such that

$$|f(h)| < \epsilon |g(h)| \quad \text{whenever} \quad |h| < \delta. \quad (1)$$

Remark 1. Equivalently, $f(h) = o(g(h))$ as $h \rightarrow 0$ if $\lim_{h \rightarrow 0} f(h)/g(h) = 0$.

Definition 2 (Big- O Notation). Let f and g be real-valued functions defined on a neighborhood of 0. The function $f(h)$ is said to be $O(g(h))$ as $h \rightarrow 0$ if $\exists M > 0$ and $\exists \delta > 0$ such that

$$|f(h)| \leq M |g(h)| \quad \text{whenever} \quad |h| < \delta. \quad (2)$$

Remark 2. By definition, any function $g(h)$ is $O(g(h))$.

Remark 3 (Interpretation of o and O). If $f(h) = o(h)$ as $h \rightarrow 0$, then $f(h)$ goes to zero *faster* than the rate of h ; if $f(h) = O(h)$ as $h \rightarrow 0$, then $f(h)$ goes to zero *at most* at the rate of h , i.e., providing the upper bound on the order.

Example 1 (Taylor Expansion). Let f be differentiable at x_0 . Then, as $h \rightarrow 0$, the first-order Taylor expansion of f around x_0 is

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(|h|). \quad (3)$$

Furthermore, if f is twice continuously differentiable in a neighborhood of x_0 , then the remainder in the first-order expansion can be sharpened to

$$f(x_0 + h) = f(x_0) + f'(x_0)h + O(|h|^2). \quad (4)$$

Remark 4. In this course, we will focus more on the first-order Taylor expansion and the $o(|h|)$ remainder term, i.e., most of the time, we just need the remainder term to go away faster than $|h|$ vanishes.

Exercise 1 (Arithmetic with o and O). Let f , g , and h be real-valued functions defined on a neighborhood of 0. Show the following as $h \rightarrow 0$:

$$\begin{aligned} o(f(h)) + o(g(h)) &= o(|f(h)| + |g(h)|), & o(f(h))o(g(h)) &= o(f(h)g(h)), \\ o(f(h)) + O(g(h)) &= O(|f(h)| + |g(h)|), & o(f(h))O(g(h)) &= o(f(h)g(h)), \\ O(f(h)) + O(g(h)) &= O(|f(h)| + |g(h)|), & O(f(h))O(g(h)) &= O(f(h)g(h)). \end{aligned}$$

Explain why having absolute values in some right-hand sides are necessary. Also convince yourself intuitively why these results are true.

B Little- o_p and Big- O_p Notation

Definition 3 (Little- o_p Notation). Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables and let $\{a_n\}_{n=1}^\infty$ be a sequence of positive constants. X_n is said to be $o_p(a_n)$ if $\forall \epsilon, \delta > 0 \exists N_{\epsilon, \delta} \in \mathbb{N}$ such that

$$\Pr \left\{ \left| \frac{X_n}{a_n} \right| > \delta \right\} < \epsilon \quad \text{whenever} \quad n > N_{\epsilon, \delta}. \quad (5)$$

Remark 5. Equivalently, $X_n = o_p(a_n)$ if $|X_n/a_n| \xrightarrow{p} 0$ as $n \rightarrow \infty$.

Definition 4 (Big- O_p Notation). Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables and let $\{a_n\}_{n=1}^\infty$ be a sequence of positive constants. X_n is said to be $O_p(a_n)$ if $\forall \epsilon > 0 \exists \delta_\epsilon > 0$ and $N_\epsilon \in \mathbb{N}$ such that

$$\Pr \left\{ \left| \frac{X_n}{a_n} \right| > \delta_\epsilon \right\} < \epsilon \quad \text{whenever} \quad n > N_\epsilon. \quad (6)$$

Remark 6. By definition, any deterministic sequence a_n is $O_p(a_n)$.

Remark 7 (Interpretation of o_p and O_p). Little- o_p means that X_n is negligible relative to a_n , i.e., X_n/a_n *converges in probability* to zero. Big- O_p means that X_n/a_n is *stochastically bounded*, i.e., the sequence X_n/a_n does not “blow up” as $n \rightarrow \infty$.

Example 2. Let $\{X_i\}_{i=1}^n$ be iid with mean zero. Then,

$$\frac{1}{n} \sum_{i=1}^n X_i = o_p(1) \quad (7)$$

by the **Weak Law of Large Numbers (WLLN)**. Furthermore, if $\text{Var}(X_i) = \sigma^2$ is finite, then by the **Central Limit Theorem (CLT)**,

$$\frac{1}{n} \sum_{i=1}^n X_i = O_p(n^{-1/2}) \quad \text{since} \quad \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i \rightsquigarrow \mathcal{N}(0, \sigma^2). \quad (8)$$

Exercise 2 (Arithmetic with o_p and O_p). Let $\{a_n\}_{n=1}^\infty$ and $\{b_n\}_{n=1}^\infty$ be sequences of positive constants. Show the following as $n \rightarrow \infty$:

$$\begin{aligned} o_p(a_n) + o_p(b_n) &= o_p(a_n + b_n), & o_p(a_n)o_p(b_n) &= o_p(a_nb_n), \\ o_p(a_n) + O_p(b_n) &= O_p(a_n + b_n), & o_p(a_n)O_p(b_n) &= o_p(a_nb_n), \\ O_p(a_n) + O_p(b_n) &= O_p(a_n + b_n), & O_p(a_n)O_p(b_n) &= O_p(a_nb_n). \end{aligned}$$

Also convince yourself intuitively why these results are true.